

Implementing Singleton pattern by creating student database which is common for all and nobody couldn't access it using constructor.

Database.cs

```
using System;
using System.Collections.Generic;

namespace CTS.SolidPrinciples.DesignPatterns
{
    public class Database
    {
        private static Database instance;
        private Dictionary<string, string> map = new Dictionary<string, string>();
        private Database()
        {
        }

        public static Database GetInstance()
        {
            if (instance == null)
            {
                instance = new Database();
            }
            return instance;
        }

        public void AddEntry(string name, string regno)
        {
            map[name] = regno;
        }

        public int NoOfStudents()
        {
            return map.Count;
        }

        public void RegNoWithName(string name)
        {
            if (map.ContainsKey(name))
            {
                Console.WriteLine(map[name]);
            }
            else
            {
                Console.WriteLine("Student not found");
            }
        }
    }
}
```

Main.cs



```
using System;

namespace CTS.SolidPrinciples.DesignPatterns
{
    class Program
    {
        static void Main(string[] args)
        {
            Database faculty = Database.GetInstance();

            Console.WriteLine(faculty.NoOfStudents());

            faculty.AddEntry("Hemanth", "231FA18278");

            Database student = Database.GetInstance();

            Console.WriteLine(student.NoOfStudents());

            student.AddEntry("Teja", "231FA18271");

            Console.WriteLine(student.NoOfStudents());

            student.RegNoWithName("Teja");
        }
    }
}
```

Output:

0

1

2

231FA18271